# ABISHEK BALU

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GitHub https://github.com/Abishek23457 Chennai, Tamil Nadu, India

Full Stack AI Application Engineer and MCA student with experience in production-grade applications using React, FastAPI, and Python. Certified Microsoft Azure Fundamentals professional and published author on AI Application Generation. Specialized in REST API design, RAG pipelines, and automated workflows.

## TECHNICAL SKILLS

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| **Languages & Core:** | Python, Java, C++, JavaScript (ES6+), TypeScript, Data Structures & Algorithms |
| **Frontend/Backend:** | React.js, Angular, HTML5/CSS3, FastAPI, REST APIs, JWT, RBAC |
| **AI / ML:** | LLM Integration (Llama 3.3), RAG Pipelines, FAISS Vector Search, Prompt Engineering |
| **Databases & Cloud:** | PostgreSQL, MongoDB, MySQL, Redis, Microsoft Azure (AZ-900), Git, Docker |
| **Tools:** | n8n Automation, Postman, VS Code, Agile/Scrum, MVC Architecture |

## WORK EXPERIENCE

### Software Development Intern
*Dec 2025 – Feb 2026*

**Calsoft Software Company Limited | India**

- Developed responsive UI components using React.js and Angular, improving page load consistency across core modules.
- Collaborated in an Agile environment (sprint planning, daily stand-ups) to deliver production-ready code via Git.
- Integrated frontend with RESTful APIs, reducing data-fetch errors by debugging JSON payloads and validating endpoints.
- Applied MVC architecture patterns, contributing to code reviews and modular system design.

## KEY PROJECTS

**BookSmart AI – Salon Booking Platform** | *FastAPI, React, PostgreSQL, Llama 3.3, FAISS, n8n*

- Built a full-stack RAG-powered chat engine using Llama 3.3 70B with FAISS vector search and Redis caching.
- Achieved sub-200ms query response times via Cohere re-ranking and hybrid search (FAISS + BM25).
- Implemented secure JWT-based RBAC and 6 modular REST API routers following clean architecture.

**Secure & Scalable Web API** | *FastAPI, PostgreSQL, SQLAlchemy, JWT*

- Architected production-grade REST API with JWT authentication and async SQLAlchemy ORM.
- Reduced API response times by ~30% through asynchronous request handling and modular design.
- Documented 15+ endpoints via OpenAPI/Swagger standards for seamless frontend-backend integration.

**StagKitchen – AI Diet & Recipe App** | *Python, React, OpenAI API*

- Developed an AI-driven planner generating personalized recipes based on ingredients and calorie targets.
- Integrated a nutrition tracking dashboard and local storage for recipe management.

## EDUCATION

### Master of Computer Applications (MCA)
*2024 – 2026 (Expected)*

**Srinivasan College of Arts and Science, Tamil Nadu**

### Bachelor of Computer Applications (BCA)
*2021 – 2024*

**Srinivasan College of Arts and Science, Tamil Nadu**

## RESEARCH & CERTIFICATIONS

- **Publication:** "The Dawn of Cinematic AI: A Study on Emerging Video Generation Models" (International Conference).
- **Certifications:** Microsoft Certified: Azure Fundamentals (AZ-900); MongoDB Basics for Students.